

All marks out of 10	Weighting
Core Simulation Features	0.3
Able to load and view scenes and interact through UI	1
Collision detection between moving spheres and other objects (spheres, cylinders, capsules, cuboids, planes)	2
Collision detection between all moving non-spherical objects	2
Collision response between moving spheres and other moving spheres with varying masses	1
Collision response including rotation (angular velocity and inertia)	2
Implementation of animated objects	1
Implementation of spawners	1
Core Concurrency Features	0.3
Peer to peer architecture (2 peers)	2
Assign ownership of objects and colour accordingly	1
Global state of simulated objects	2
Ease of network setup	1
Network performance	2
Graphics threads (main thread)	2
Networking threads	2
Simulation threads	2
Frequency adjustment of graphics	1
Frequency adjustment of networking	1
Frequency adjustment of simulation	1
Processor affinity	1
Advanced Simulation Features	0.1
Implementation of chosen advanced simulation feature	3
Debugging visualisation and UI interaction	1
Evidence of own independent research in report	1
Advanced Concurrency Features	0.1
Peer to peer architecture (4 peers)	1
Correction algorithms to fix errors in shape position/orientation, e.g. interpolation, extrapolation.	2
Robustness of networking due to packet loss and/or latency	2
Extended Simulation Features	0.1
Implementation of chosen advanced simulation feature	3
Debugging visualisation and UI interaction	1
Evidence of own independent research in report	1
Extended Concurrency Features	0.1
Elements of the simulation implemented on the GPU using Compute Shaders	1
Final Lab Total	1
Lab book (marked out of 3)	0.2
Lab 1 - SJG	3
Lab 2 - SJG	3
Lab 3 - SJG	3
Lab 4 - SJG	3
Lab 5 - SJG	3
Lab 6 - SJG	3
Lab A - WJV	3
Lab B - WJV	3
Lab C - WJV	3
Lab D - WJV	3
Lab E - WJV	3
Lab F - WJV	3
Best 4 marks from SJG Labs	12
Best 4 marks from WJV Labs	12
Lab I (final) - PBM	3
Lab I (final) - networking and concurrency	3
Deliverable (marked out of 10)	0.05
Video	1
Quality of demonstration	1
Final Lab Total	0.75
Lab book (marked out of 3)	0.2
Deliverable (marked out of 10)	0.05
Total	1